Bayesian quantile correction and synthesis for climate products

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QUESTION
Correction before synthesis

How can historical archives estimate source corrections before forecasts with different horizons are blended into one predictive distribution?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge, modeled on the $\log(1+x)$ scale with x in m^3/s. Before each forecast origin, USGS observations and retrospective ECMWF/-GloFAS and NOAA/NWS products identify time-varying source corrections. After that origin, issued ensembles and exogenous information produce source-adjusted predictive flow quantiles.

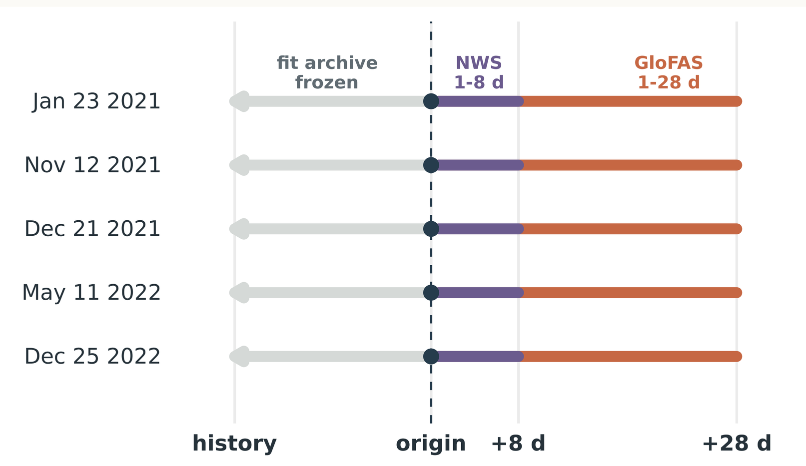
USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.

ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION
Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE
Scalable Bayesian quantile fitting

Seven independent Dynamic Quantile Linear Model (DQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7	$\leq$12,995	~2–4 h
quantile lanes	USGS archive days	fit + synthesis

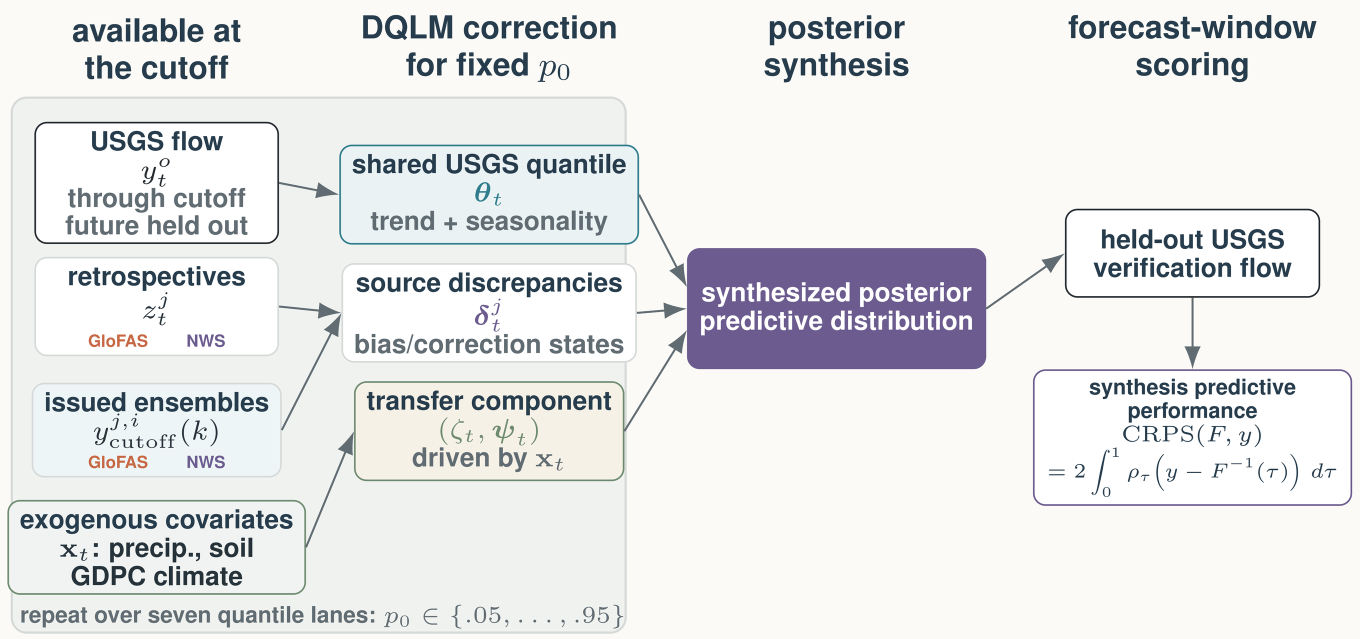
Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

State distribution updates
Mean-field variational Bayes (MFVB) uses Kalman filtering and backward smoothing equations to update Gaussian state distributions within each latent-state block.

Non-conjugate updates
Variational Bayes with Laplace–Delta (VB-LD) uses local modes, curvature, and Delta-method moments for non-conjugate extended asymmetric Laplace (exAL) scale and skewness terms.

State evolution
Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate adjustments.

MODEL
Main Workflow



Source-aware DQLM workflow. For each target quantile p_0, pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

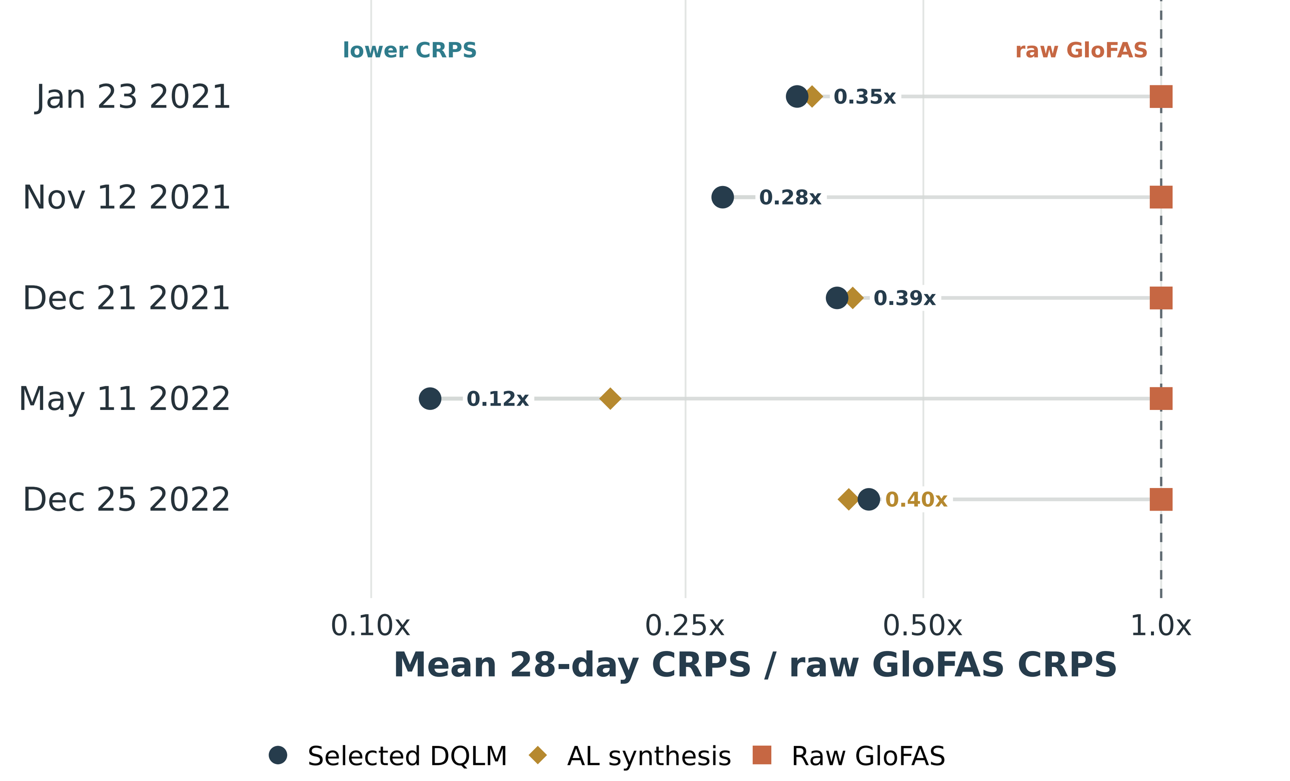
Dynamic Quantile Linear Model
For each target quantile p_0, let T denote the forecast origin/cutoff. The Dynamic Quantile Linear Model (DQLM) links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Product-family discrepancies learn source corrections before the origin; the same source channels, with forecast-window covariates $\mathbf{x}_{T+k}$, then define corrected predictive quantiles after the origin.

$$
\begin{aligned}
&\text{Observed measurements:} && y_t^o && \sim \mathrm{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o), && 1 \le t \le T,\\
&\text{Retrospective products:} && z_t^j && \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_t'(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j\right), && j \in \mathcal{J},\ 1 \le t \le T,\\
&\text{Forecast products:} && y_T^{j,i}(k) && \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j\right), && j \in \mathcal{J},\ i \in \mathcal{I}_j,\ k \in \mathcal{K}_j,\\
&\text{Trend and seasonal states:} && \boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} && \sim \mathcal{N}\left(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t^{\theta}\right), && 1 \le t \le T + K_{\max},\\
&\text{Source correction states:} && \boldsymbol{\delta}_t^j \mid \boldsymbol{\delta}_{t-1}^j && \sim \mathcal{N}\left(\mathbf{G}_t\boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta j}\right), && j \in \mathcal{J},\ 1 \le t \le T + K_{\max},\\
&\text{Exogenous adjustment:} && \zeta_t \mid \zeta_{t-1}, \boldsymbol{\psi}_{t-1} && \sim \mathcal{N}\left(\lambda\zeta_{t-1} + \mathbf{x}_t'\boldsymbol{\psi}_{t-1}, w_t^{\zeta}\right), && 1 \le t \le T + K_{\max},\\
&\text{Covariate-adjustment states:} && \boldsymbol{\psi}_t \mid \boldsymbol{\psi}_{t-1} && \sim \mathcal{N}\left(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^{\psi}\right), && 1 \le t \le T + K_{\max}.
\end{aligned}
$$

exAL_{p_0} is the extended asymmetric Laplace working likelihood for target quantile p_0; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is $\boldsymbol{\theta}_t$; source discrepancies/corrections are $\boldsymbol{\delta}_t^j$; and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-adjustment block driven by covariates $\mathbf{x}_t$ such as precipitation, soil moisture, and the GDPC climate-index summary. Component discounting controls the evolution variances.

RESULTS
28-day CRPS comparison

Selected DQLM has the lowest 28-day CRPS at four origins; AL synthesis leads once CRPS ↓

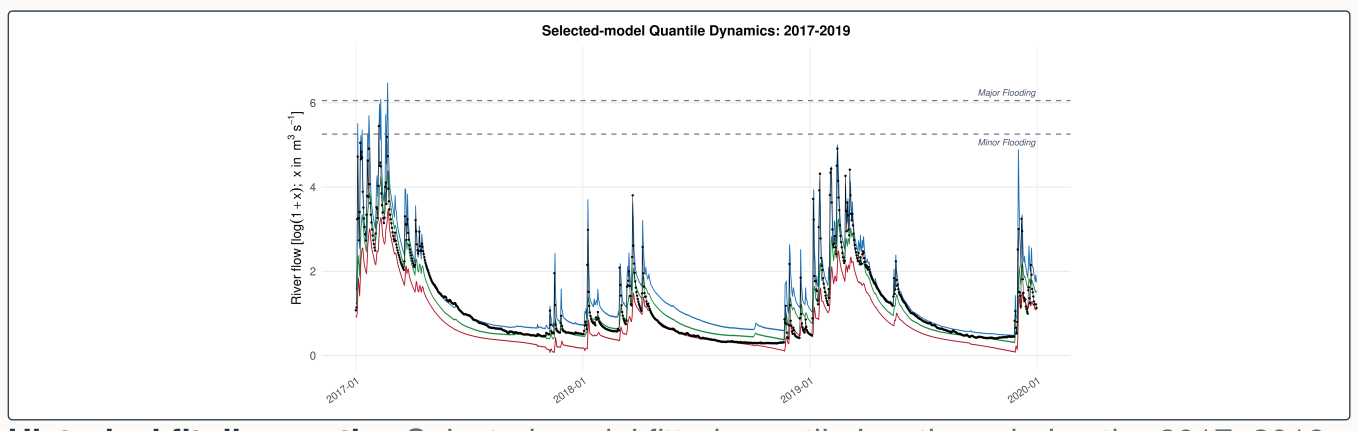


Exception: AL synthesis has the lowest 28-day CRPS at the 2022-12-25 origin.

How to read this panel. Each point is mean forecast-window CRPS divided by raw GloFAS CRPS at the same origin. Values below 1.0 improve on the uncorrected 28-day GloFAS benchmark.

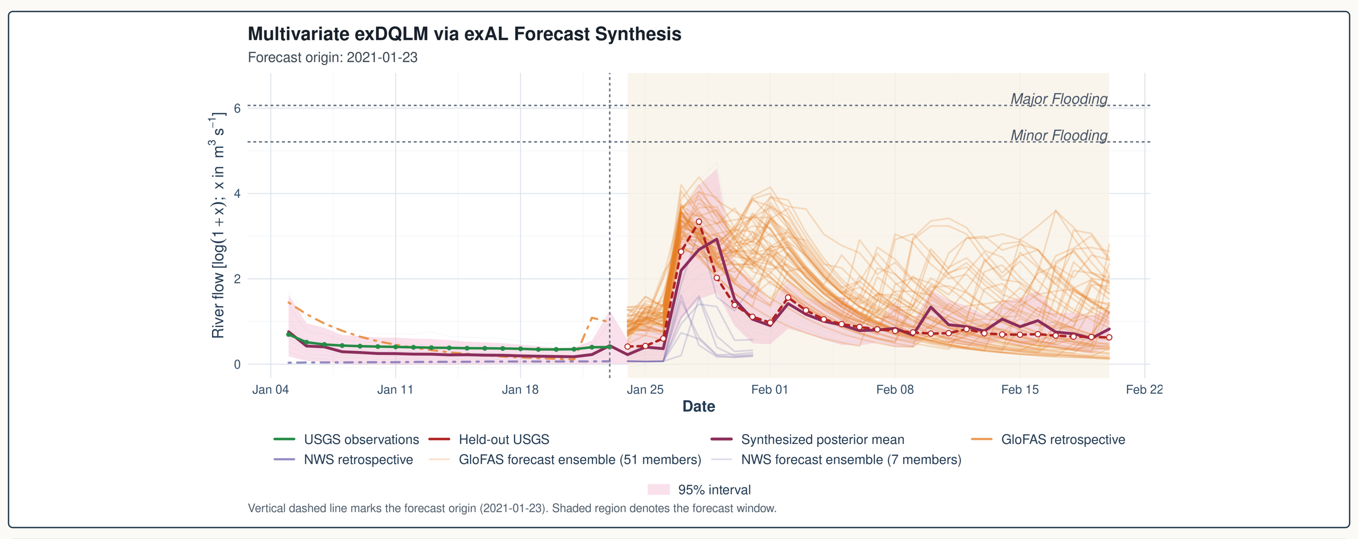
Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin; raw NWS is evaluated over common leads 1–8 in the horizon-matched tables, not mixed into this 28-day GloFAS-supported benchmark.

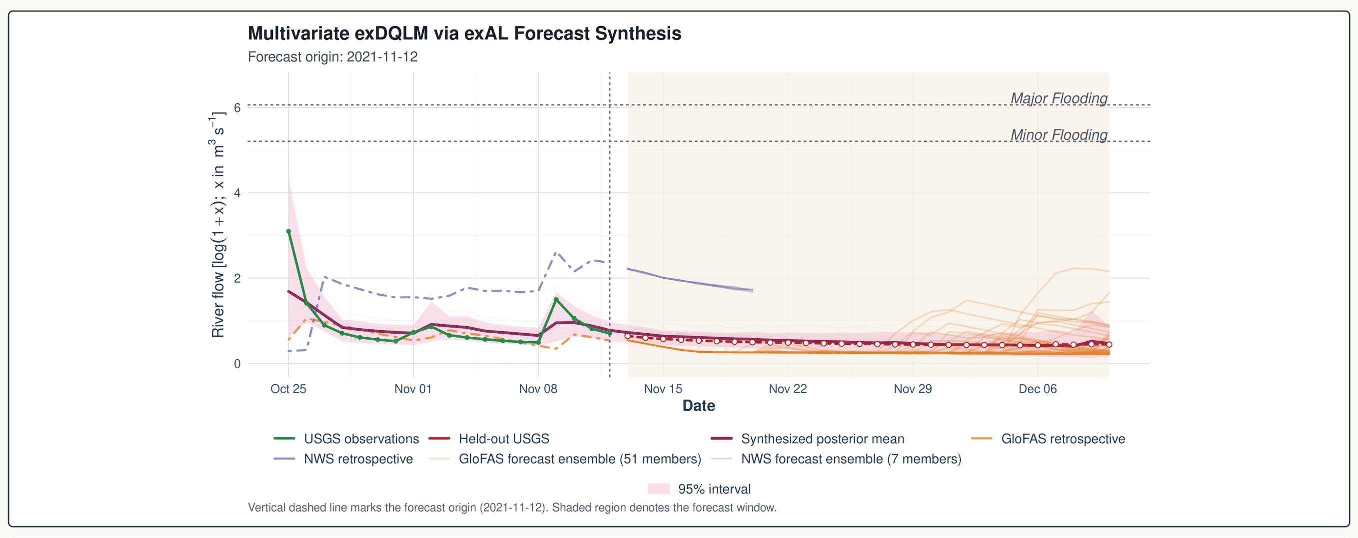
FIT
Wet-period quantile fit

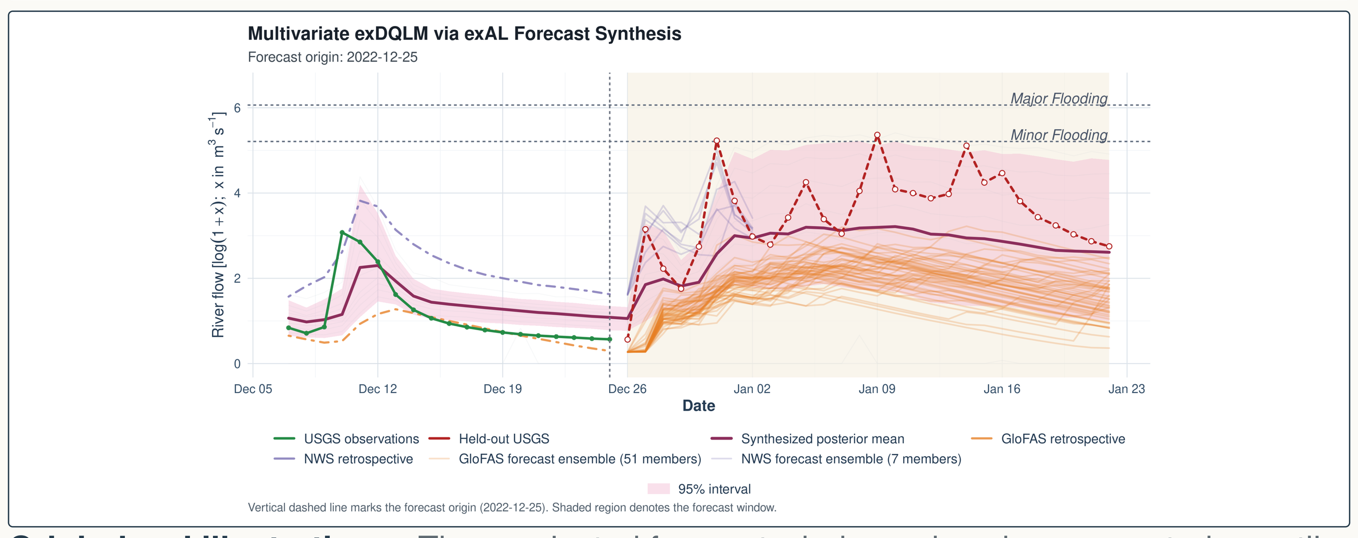


Historical fit diagnostic. Selected-model fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE
Three forecast-window examples

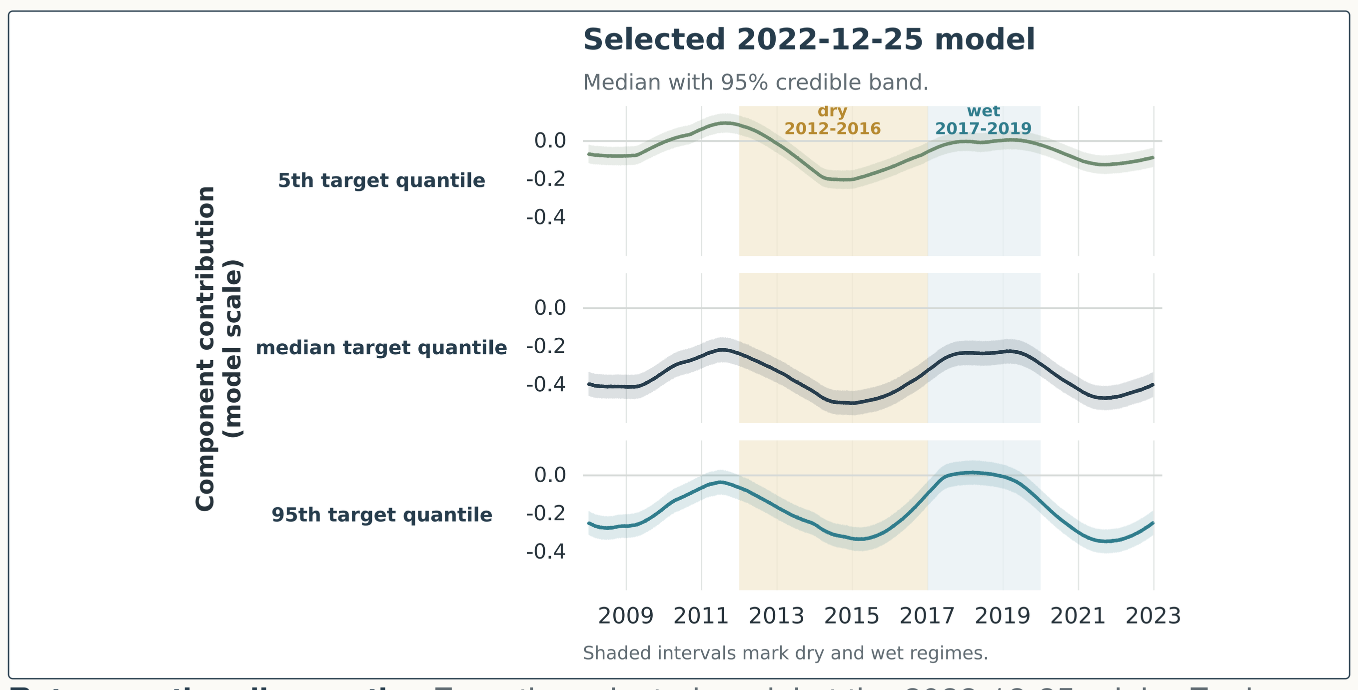






Origin-level illustrations. Three selected forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS
80-month seasonal component



Retrospective diagnostic. From the selected model at the 2022-12-25 origin. Each row shows the posterior median and 95% credible band for one target-quantile lane over 2008–2022; shaded intervals mark dry and wet regimes.

TAKEAWAYS
Main takeaways

- Historical archives learn source-specific corrections before forecast products are blended.
- Seven corrected quantile lanes are synthesized into one posterior predictive distribution.
- In this five-origin study, the selected DQLM has the lowest 28-day CRPS at four origins; 2022-12-25 favors AL synthesis.
- Fitted quantile dynamics remain interpretable: spread widens during wet high-flow periods.

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Code and reproducibility
github.com/AntonioAPDL/Project1